

THE PROBLEM

The Riesel Problem — Base 71

What is a Riesel number?

An odd k such that $k \cdot 2^n - 1$ is composite for EVERY $n \geq 1$.
Hans Riesel (1956): 509203 is one — via a covering set of primes.
Conjecture: it is the smallest. Still unproven today.

A 70-year lineage

- 1956** Riesel: $k \cdot 2^n - 1$ problem posed (43 k -values still open)
- 1960** Sierpiński: $k \cdot 2^n + 1$ analogue (78557 conjectured smallest)
- 2007** CRUS project: prove the conjecture for EVERY base b
- 2023** Base 71 opened (Nov 24): $k \cdot 71^n - 1$ for all k below CK
- 2025** Status: $k > 1M$ tested only to $n=1000$ — 19,068,114 k 's open

THE OPEN QUESTION (one per unproven k)

Does $k \cdot 71^n - 1$ ever produce a prime?

19,068,114

k -values unproven
(on the CRUS list)

$n \leq 1000$

how far $k > 1M$ had
been tested (Oct 2025)

unreserved

open to anyone
(status: Jul 22, 2026)

Why one prime settles one k

To prove the base-71 conjecture, every k below the conjectured CK must yield a prime $k \cdot 71^n - 1$.
Each unproven k is its own open question — and a SINGLE prime ends it forever.
Bonus: $N+1 = k \cdot 71^n$ is fully factored, so each prime found admits a rigorous
Brillhart-Lehmer-Selfridge (Lucas $N+1$) primality proof — no probability anywhere.

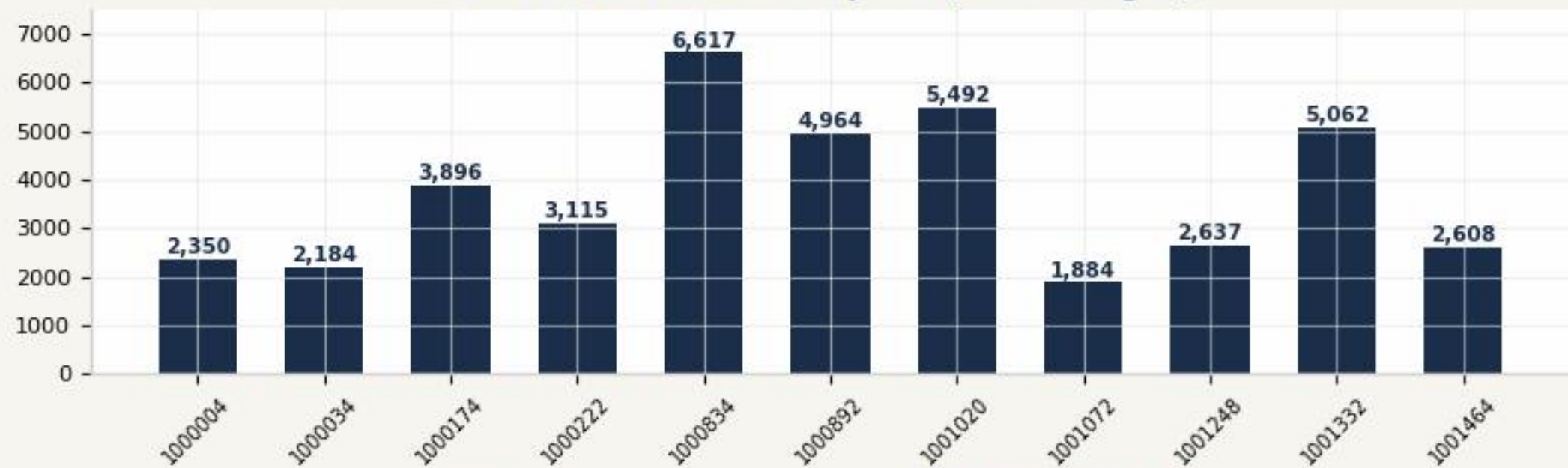
THE SOLVED ANSWERS

11 Open Instances — Settled & Proven

Least n such that $k \cdot 71^n - 1$ is prime (all smaller n provably composite)

k	least n	digits of the prime	Lucas D
1000004	1266	2,350	5
1000034	1176	2,184	-19
1000174	2101	3,896	-19
1000222	1679	3,115	53
1000834	3571	6,617	61
1000892	2678	4,964	21
1001020	2963	5,492	21
1001072	1014	1,884	65
1001248	1421	2,637	5
1001332	2731	5,062	37
1001464	1405	2,608	5

Size of each discovered prime (decimal digits)



Why these answers are PROOF-GRADE

1. SIEVE
n with a factor $\leq 3 \times 10^5$
→ provably composite

2. FERMAT
base-2 witness
→ provably composite

3. BPSW × 2
GMP + sympy agree
→ prime

4. LUCAS N+1
BLS proof, $N+1=k \cdot 71^n$
fully factored → PRIME

11 of the first 23 remaining k 's settled (48%); 12 more pushed from $n=1000$ to $n=4000$ (still open).
Each row removes one k from the CRUS Riesel base-71 list — 19,068,114 remained as of Oct 2025.
Riesel problem 1956 · CRUS project 2007 · base 71 opened 2023 · settled 2026-07-24